

Class 11: AlphaFold

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```
library(bio3d)
```

```
pdb <- read.pdb("hivpr_dimer_23119/hivpr_dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_0.pdb")
```

Make a vector of input PDB file names that can read into R

```
pdbfiles <- list.files("hivpr_dimer_23119/",  
                      pattern = ".pdb",  
                      full.names = TRUE)
```

```
library(bio3d)
```

```
# Read all data from Models  
# and superpose/fit coords  
pdbb <- pdbaln(pdbfiles, fit=TRUE, exefile="msa")
```

Reading PDB files:

```
hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_0.pdb  
hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_0.pdb  
hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_0.pdb  
hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_3_seed_0.pdb  
hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_2_seed_0.pdb  
.....
```

Extracting sequences

```
pdb/seq: 1   name: hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_0.pdb  
pdb/seq: 2   name: hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_0.pdb  
pdb/seq: 3   name: hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_0.pdb  
pdb/seq: 4   name: hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_3_seed_0.pdb  
pdb/seq: 5   name: hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_2_seed_0.pdb
```

```
rd <- rmsd(pdbbs, fit=T)
```

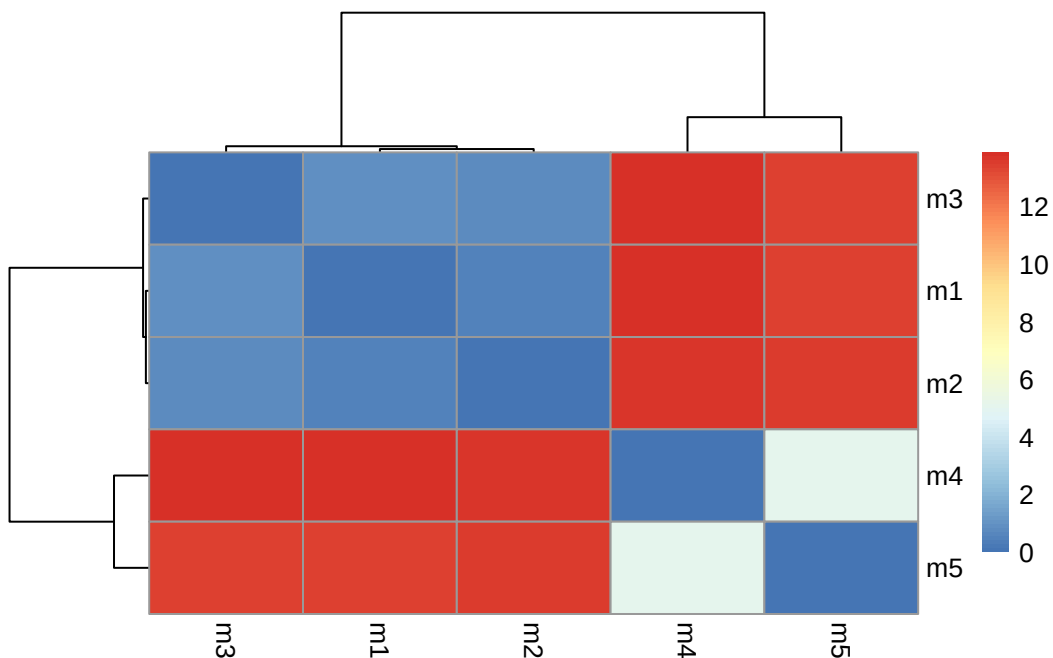
Warning in rmsd(pdbbs, fit = T): No indices provided, using the 198 non NA positions

```
range(rd)
```

```
[1] 0.000 13.888
```

```
library(pheatmap)
```

```
colnames(rd) <- paste0("m",1:5)  
rownames(rd) <- paste0("m",1:5)  
pheatmap(rd)
```



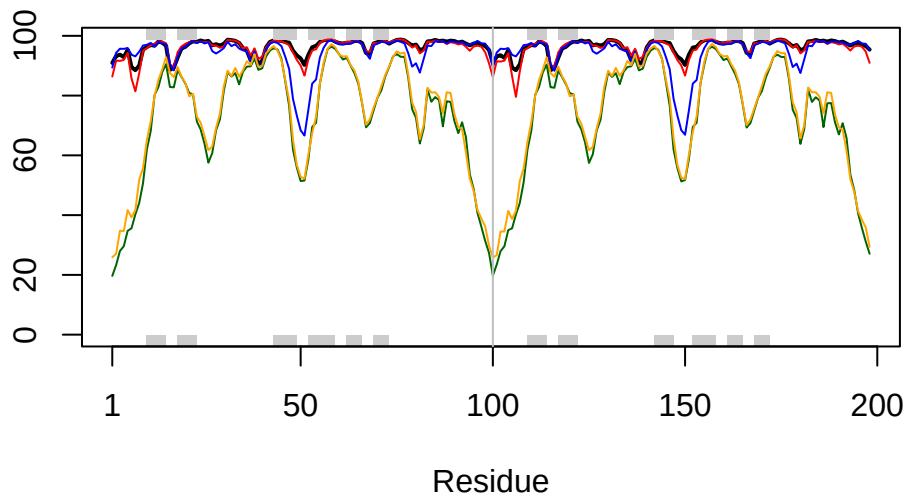
```
# Read a reference PDB structure  
pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```

plotb3(pdbb$b[1,], typ="l", lwd=2, sse=pdb)
points(pdbb$b[2,], typ="l", col="red")
points(pdbb$b[3,], typ="l", col="blue")
points(pdbb$b[4,], typ="l", col="darkgreen")
points(pdbb$b[5,], typ="l", col="orange")
abline(v=100, col="gray")

```



```

core <- core.find(pdbb)

```

```

core size 197 of 198  vol = 6732.277
core size 196 of 198  vol = 6404.498
core size 195 of 198  vol = 5109.231
core size 194 of 198  vol = 3180.814
core size 193 of 198  vol = 2542.63
core size 192 of 198  vol = 2361.481
core size 191 of 198  vol = 2245.469
core size 190 of 198  vol = 2128.99
core size 189 of 198  vol = 2041.545
core size 188 of 198  vol = 1956.704
core size 187 of 198  vol = 1885.422
core size 186 of 198  vol = 1814.183
core size 185 of 198  vol = 1753.586

```

core size 184 of 198	vol = 1705.055
core size 183 of 198	vol = 1642.105
core size 182 of 198	vol = 1599.234
core size 181 of 198	vol = 1544.183
core size 180 of 198	vol = 1499.919
core size 179 of 198	vol = 1453.589
core size 178 of 198	vol = 1411.37
core size 177 of 198	vol = 1374.172
core size 176 of 198	vol = 1348.898
core size 175 of 198	vol = 1322.722
core size 174 of 198	vol = 1278.394
core size 173 of 198	vol = 1237.868
core size 172 of 198	vol = 1194.871
core size 171 of 198	vol = 1149.916
core size 170 of 198	vol = 1115.748
core size 169 of 198	vol = 1074.461
core size 168 of 198	vol = 1041.82
core size 167 of 198	vol = 1015.099
core size 166 of 198	vol = 982.853
core size 165 of 198	vol = 948.684
core size 164 of 198	vol = 910.883
core size 163 of 198	vol = 877.107
core size 162 of 198	vol = 848.809
core size 161 of 198	vol = 826.047
core size 160 of 198	vol = 798.374
core size 159 of 198	vol = 775.687
core size 158 of 198	vol = 748.519
core size 157 of 198	vol = 729.604
core size 156 of 198	vol = 703.151
core size 155 of 198	vol = 676.43
core size 154 of 198	vol = 656.323
core size 153 of 198	vol = 631.851
core size 152 of 198	vol = 605.743
core size 151 of 198	vol = 569.843
core size 150 of 198	vol = 544.115
core size 149 of 198	vol = 517.354
core size 148 of 198	vol = 494.921
core size 147 of 198	vol = 479.374
core size 146 of 198	vol = 453.303
core size 145 of 198	vol = 435.934
core size 144 of 198	vol = 417.06
core size 143 of 198	vol = 398.538
core size 142 of 198	vol = 374.827

core size 141 of 198	vol = 354.269
core size 140 of 198	vol = 336.307
core size 139 of 198	vol = 318.544
core size 138 of 198	vol = 308.804
core size 137 of 198	vol = 295.052
core size 136 of 198	vol = 279.398
core size 135 of 198	vol = 269.454
core size 134 of 198	vol = 254.195
core size 133 of 198	vol = 239.592
core size 132 of 198	vol = 224.085
core size 131 of 198	vol = 211.153
core size 130 of 198	vol = 200.038
core size 129 of 198	vol = 189.628
core size 128 of 198	vol = 179.866
core size 127 of 198	vol = 170.227
core size 126 of 198	vol = 163.445
core size 125 of 198	vol = 154.14
core size 124 of 198	vol = 147.598
core size 123 of 198	vol = 141.253
core size 122 of 198	vol = 135.249
core size 121 of 198	vol = 128.059
core size 120 of 198	vol = 120.252
core size 119 of 198	vol = 112.759
core size 118 of 198	vol = 105.287
core size 117 of 198	vol = 99.903
core size 116 of 198	vol = 94.13
core size 115 of 198	vol = 87.053
core size 114 of 198	vol = 81.136
core size 113 of 198	vol = 74.821
core size 112 of 198	vol = 67.365
core size 111 of 198	vol = 59.58
core size 110 of 198	vol = 53.042
core size 109 of 198	vol = 46.554
core size 108 of 198	vol = 43.411
core size 107 of 198	vol = 40.091
core size 106 of 198	vol = 34.351
core size 105 of 198	vol = 31.465
core size 104 of 198	vol = 27.543
core size 103 of 198	vol = 24.482
core size 102 of 198	vol = 21.186
core size 101 of 198	vol = 18.171
core size 100 of 198	vol = 17.001
core size 99 of 198	vol = 12.927

```

core size 98 of 198  vol = 11.103
core size 97 of 198  vol = 9.485
core size 96 of 198  vol = 7.712
core size 95 of 198  vol = 6.546
core size 94 of 198  vol = 5.671
core size 93 of 198  vol = 4.844
core size 92 of 198  vol = 3.497
core size 91 of 198  vol = 2.256
core size 90 of 198  vol = 1.684
core size 89 of 198  vol = 1.116
core size 88 of 198  vol = 0.972
core size 87 of 198  vol = 0.91
core size 86 of 198  vol = 0.723
core size 85 of 198  vol = 0.563
core size 84 of 198  vol = 0.458
FINISHED: Min vol ( 0.5 ) reached

```

```
core.inds <- print(core, vol=0.5)
```

```
# 85 positions (cumulative volume <= 0.5 Angstrom^3)
```

```

start end length
1      9 48      40
2     52 96      45

```

```

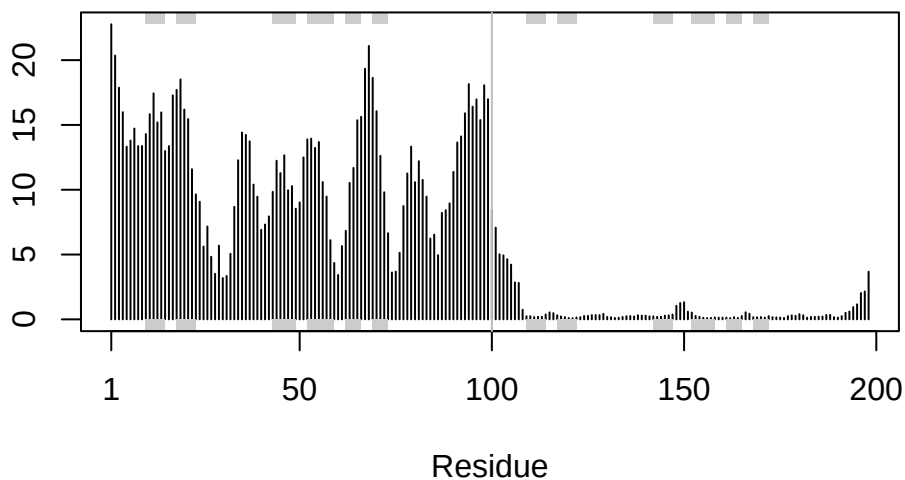
xyz <- pdbfit(pdb, core.inds, outpath="corefit_structures")
rf <- rmsf(xyz)

```

```

plotb3(rf, sse=pdb)
abline(v=100, col="gray", ylab="RMSF")

```



```
library(jsonlite)
```

```
# Listing of all PAE JSON files
```

```
pae_files <- list.files(path="hivpr_dimer_23119",
                        pattern=".*model.*\\.json",
                        full.names = TRUE)
```

```
pae1 <- read_json(pae_files[1],simplifyVector = TRUE)
```

```
pae5 <- read_json(pae_files[5],simplifyVector = TRUE)
```

```
attributes(pae1)
```

```
$names
```

```
[1] "plddt" "max_pae" "pae" "ptm" "iptm"
```

```
# Per-residue pLDDT scores
```

```
# same as B-factor of PDB..
```

```
head(pae1$plddt)
```

```
[1] 90.94 93.25 93.69 92.88 95.31 89.50
```

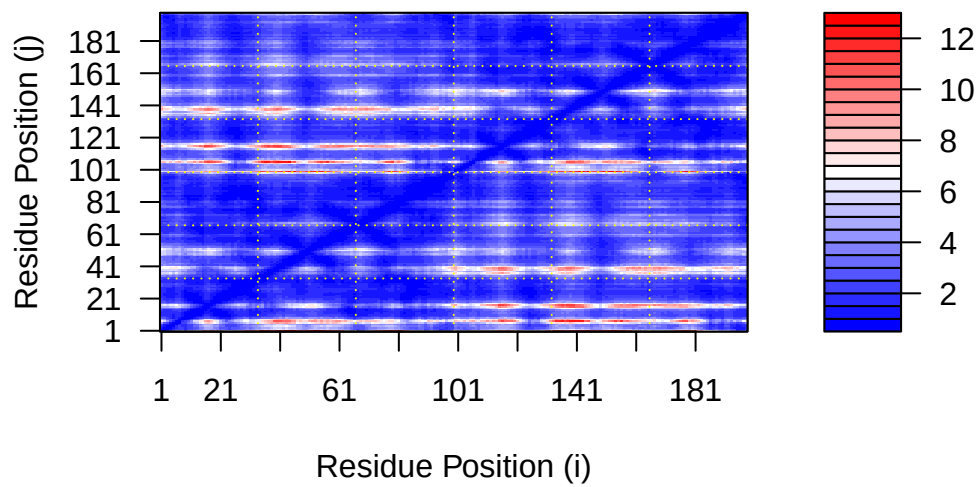
```
pae1$max_pae
```

```
[1] 12.6875
```

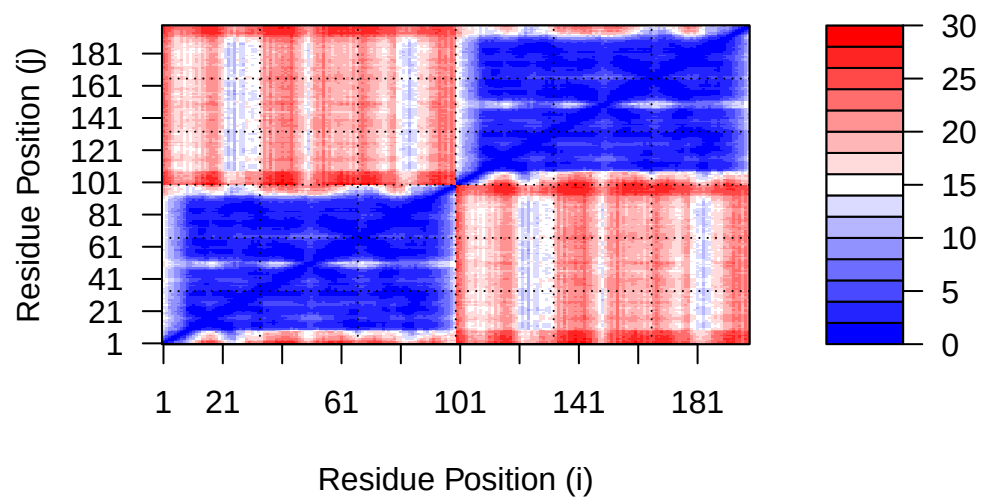
```
pae5$max_pae
```

```
[1] 29.54688
```

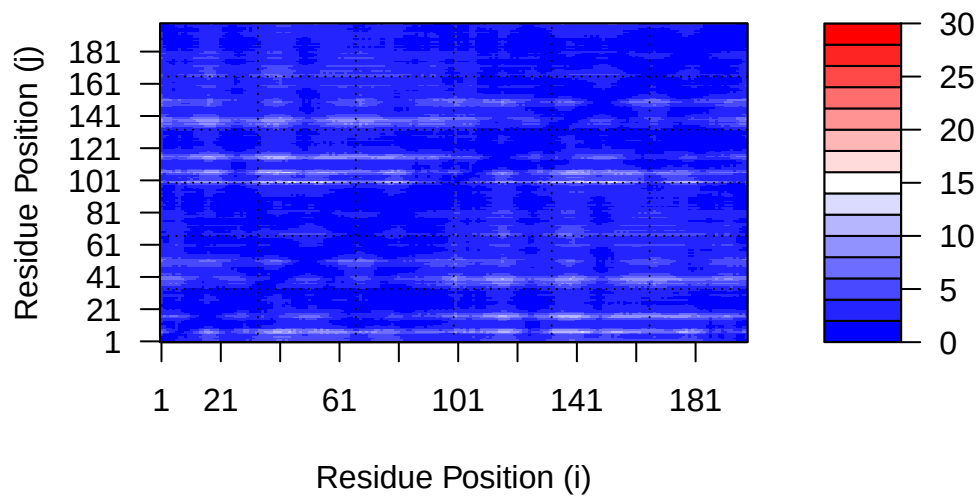
```
plot.dmat(pae1$pae,  
          xlab="Residue Position (i)",  
          ylab="Residue Position (j)")
```



```
plot.dmat(pae5$pae,  
          xlab="Residue Position (i)",  
          ylab="Residue Position (j)",  
          grid.col = "black",  
          zlim=c(0,30))
```

```
plot.dmat(pae1$paes,
  xlab="Residue Position (i)",
  ylab="Residue Position (j)",
  grid.col = "black",
  zlim=c(0,30))
```



```
aln_file <- list.files(path="hivpr_dimer_23119",
                      pattern=".a3m$",
                      full.names = TRUE)
aln_file
```

```
[1] "hivpr_dimer_23119/hivpr_dimer_23119.a3m"
```

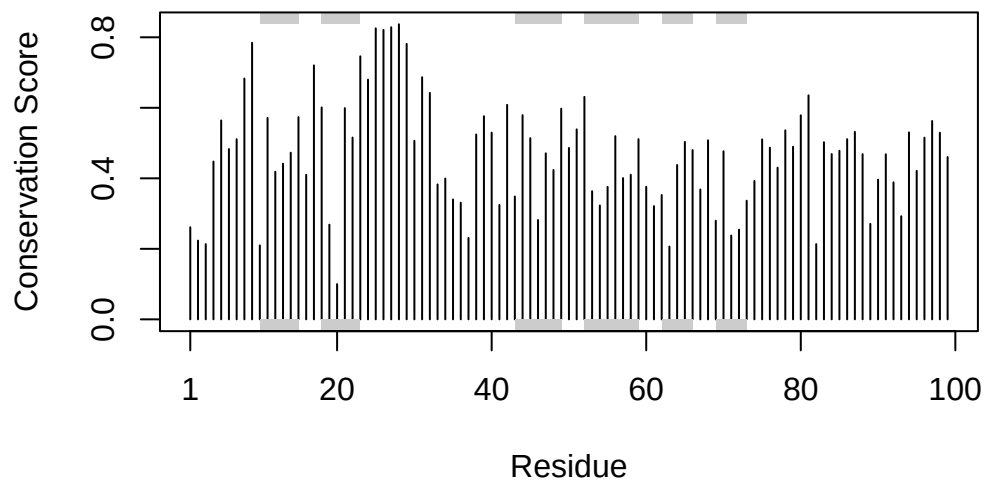
```
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

```
[1] " ** Duplicated sequence id's: 101 **"
[2] " ** Duplicated sequence id's: 101 **"
```

```
dim(aln$ali)
```

```
[1] 5397 132
```

```
sim <- conserv(aln)
plotb3(sim[1:99], sse=trim.pdb(pdb, chain="A"),
       ylab="Conservation Score")
```



```
con <- consensus(aln, cutoff = 0.9)
con$seq
```

```
[1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-" "-"
[37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```

```
m1.pdb <- read.pdb(pdbfiles[1])
occ <- vec2resno(c(sim[1:99], sim[1:99]), m1.pdb$atom$resno)
write.pdb(m1.pdb, o=occ, file="m1_conserv.pdb")
```